



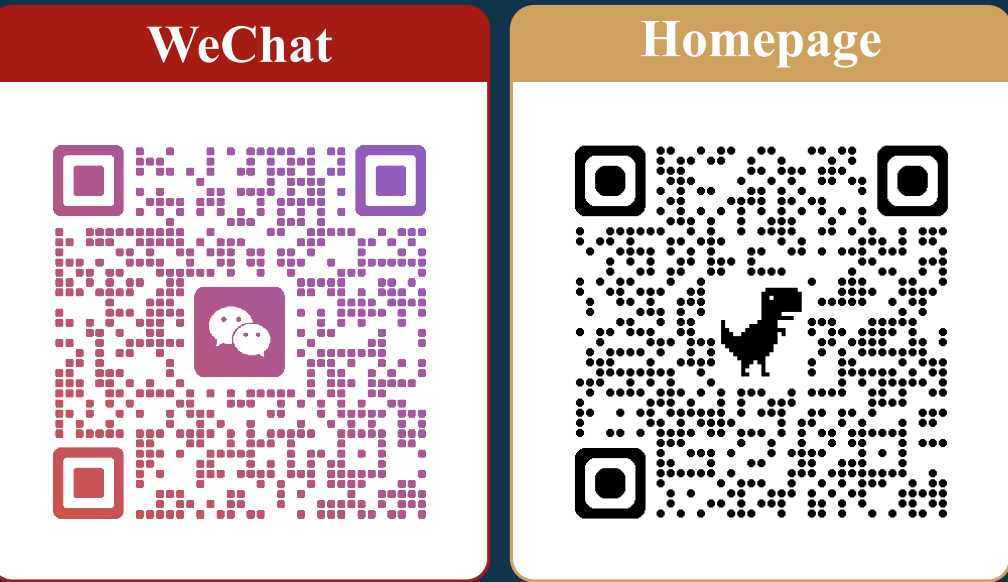
CWF: A Collaborative Writing Framework for Personalized and Reliable Popular Science Writing



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Introduction and evaluation

Popular science writing must preserve scientific accuracy while matching readers' knowledge, cognitive capacity and interests. Simplifying technical content can distort facts, whereas factual revision can make prose less accessible. CWF treats **reader adaptation and factual reliability as a joint writing task**, rather than optimizing either objective in isolation. Supervised adaptation needs costly audience-domain paired data; retrieval-based fact-checking struggles with missing or conflicting evidence. Reusable experts and multi-perspective verification address these limitations.

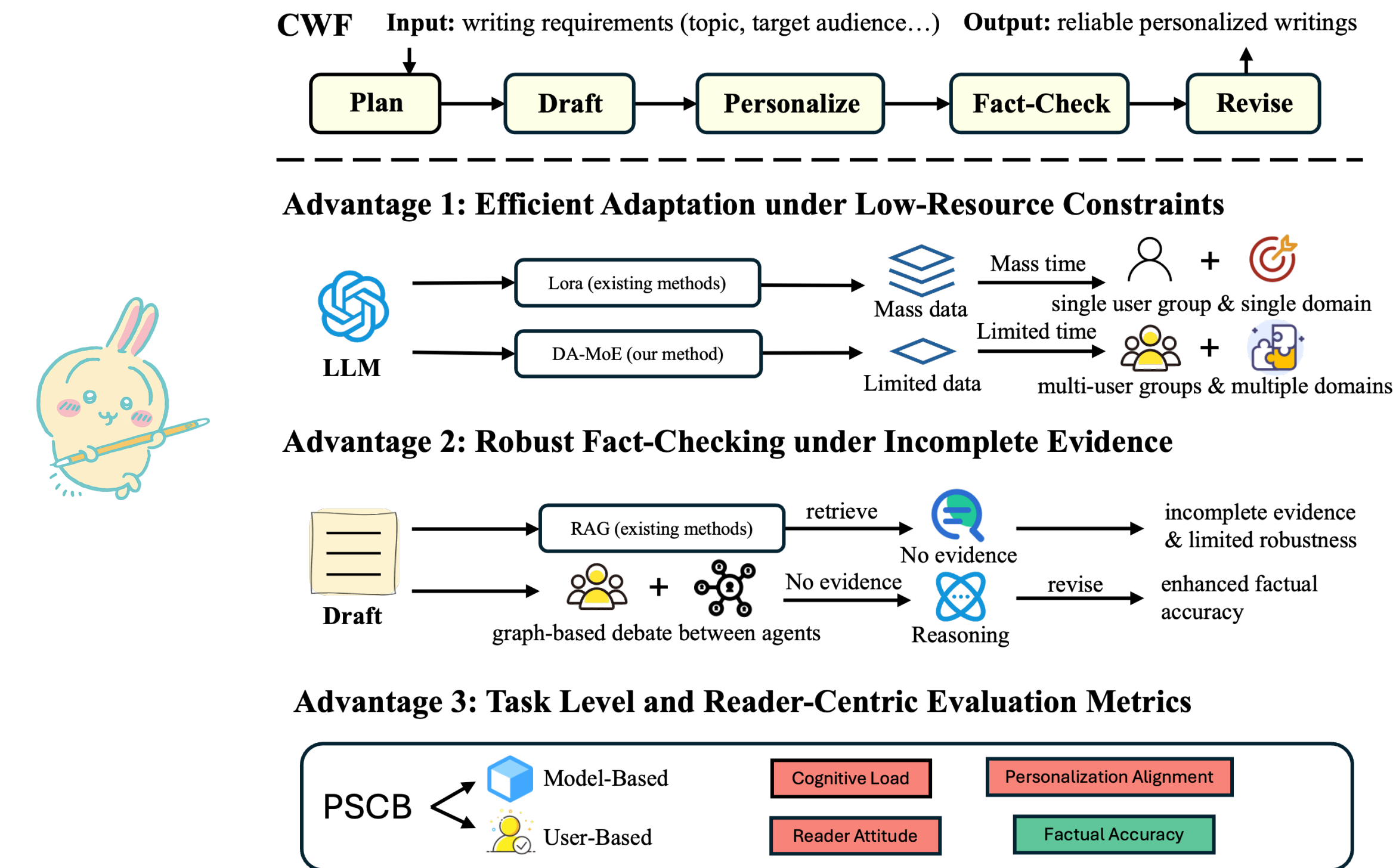


Figure 1. CWF versus specialized fine-tuning and RAG approaches (original, p. 1).

Three contributions. (1) PSCB jointly evaluates audience adaptation and factual reliability. (2) DA-MoE decouples audience expression from scientific expertise, reducing reliance on paired training data. (3) Multi-agent debate and graph inference produce traceable claim confidence and guide revision under incomplete evidence.

PSCB: reader-centered evaluation

39,134 training entries span AI, Biology and Medicine for child, teen and adult readers. Texts come from science-oriented WeChat accounts and books. Separate audience and domain data support expert specialization without requiring every entry to cover both attributes.

Ten dimensions, three metrics. Cognitive Load (CL) measures intrinsic fit, extraneous burden control and germane support [3]. Personalization Alignment (PA) measures relevance, knowledge-level fit, style consistency and contextual resonance. Reader Attitude (RA) measures engagement, trust and continuance intention. Dimension scores range from 0 to 5.

A meta-evaluator assigns dimension weights from only the topic and persona; a judge scores each dimension with a rationale. Weights are nonnegative and sum to one. CL, PA and RA are equally averaged:

$$S_{i,m} = \sum_{d \in D_m} w_{i,d}^{(m)} s_{i,d}, \quad \sum_d w_{i,d}^{(m)} = 1.$$

Human protocol. A 40-item questionnaire retains 90 readers (30 per group), 540 of 585 responses and six anonymized systems. Counterbalanced presentation, reverse coding and response-quality screening support comparability. Qwen-Plus dynamic weighting reaches MAE 0.417, Spearman $\rho = 0.829$ and 86.7% pairwise ordering accuracy (Figure 3); inter-participant agreement is 0.781–0.803.

Efficiency through expert reuse

Method	Training Time	Data Amount	Data Type	Average PSCB score
LoRA	2.1 hrs	5000	Dual-style	2.985
MoE	8.3 hrs	20000	Single-style	3.059
DA-MoE	22 min	1000	Single-style	3.451

Table 2. Complete training-time, data and PSCB comparison (original, p. 5).

22 minutes with 1,000 single-style examples describes lightweight adaptation that reuses existing experts, excluding their initial pretraining. One V100 (32 GB) is used; DA-MoE scores 3.451 versus LoRA 2.985 and MoE 3.059.

Framework and method

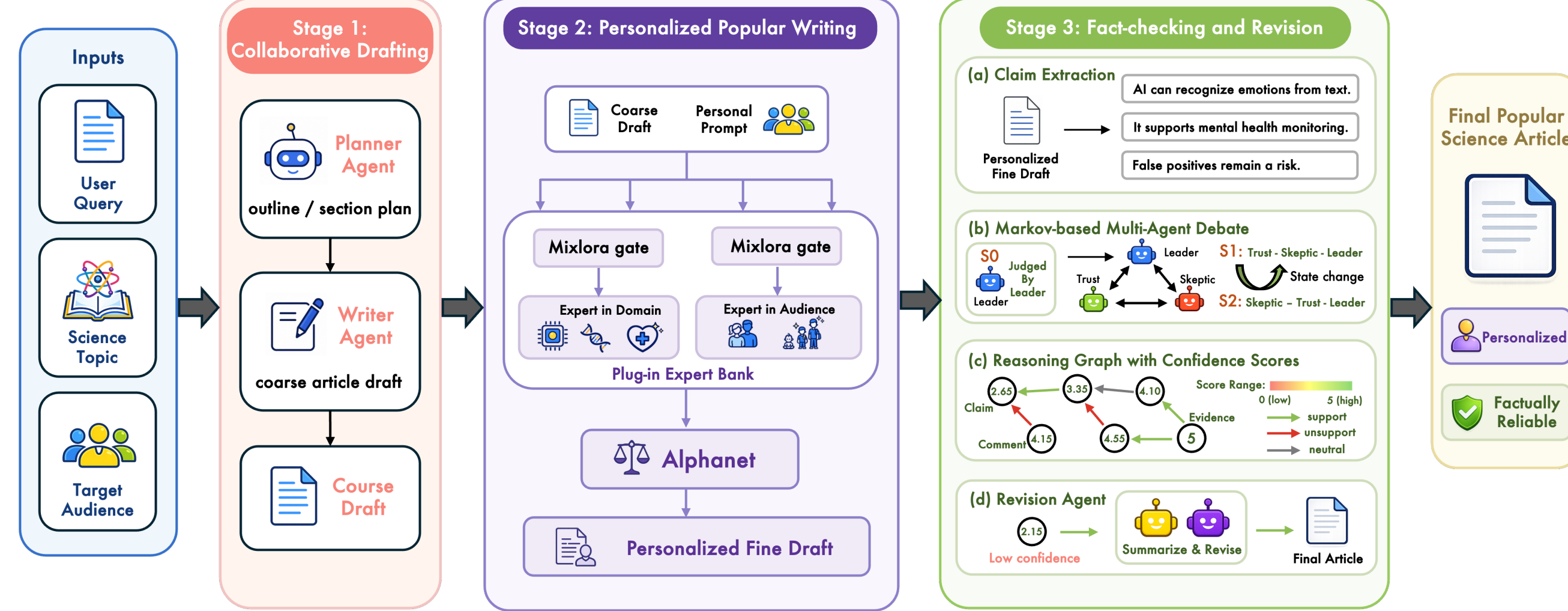


Figure 2. Complete pipeline: collaborative drafting, DA-MoE personalization, and multi-agent verification with graph-based revision (original, p. 4).

1. Collaborative drafting: organize before expanding

A planner develops the article structure and section summaries. A writer expands them into coherent passages with examples and transitions. This separation improves narrative coherence.

2. DA-MoE: compose audience and knowledge experts

Two MoE networks use Qwen2.5-3B experts: audience experts specialize in reader-oriented expression, while knowledge experts specialize in scientific domains. MixLoRA [1] supplies plug-and-play routing. Each expert is independently fine-tuned on its assigned data; experts are then frozen while lightweight gates learn input-dependent weights. For $r \in \{s, k\}$ (audience, knowledge):

$$g_r(x) = \text{softmax}(\text{MLP}_r(x)),$$

$$h_r(x) = \sum_{j=1}^T g_j^r(x) E_j^r(x).$$

Reusable modules allow new experts without full-model retraining.

3. AlphaNet: balance style and domain content

An MLP predicts the input-dependent fusion coefficient. The final hidden representation combines the two networks:

$$\alpha = \text{MLP}_{\text{alphanet}}(x),$$

$$h(x) = \alpha h_s(x) + (1 - \alpha) h_k(x).$$

Mean audience weights are 0.686 for children, 0.262 for teens and 0.183 for adults (100 samples each), indicating greater emphasis on reader-oriented expression for younger readers.

4. Verify atomic claims through role-specific debate

Paragraphs are decomposed into independently verifiable claims [2,4]; coverage weights sum to one and support confidence aggregation. **Leader** makes the scientific judgment, **Trust** searches for supporting evidence and complementary explanations, and **Skeptic** challenges unsupported claims and weak evidence. A supportive Leader invokes Skeptic first; a neutral or opposing Leader invokes Trust first. The latest Leader judgment controls the next debate order:

$$S_{t+1} = \begin{cases} S_2, & f(L_t) = 1, \\ S_1, & f(L_t) = 0. \end{cases}$$

Here S_1 follows Leader–Trust–Skeptic and S_2 follows Leader–Skeptic–Trust; $f(L_t)$ distinguishes supportive from neutral or opposing stances. Leader updates its judgment after both agents respond. Debate stops at a shared stance after a full round or at three rounds.

5. Propagate confidence, then revise

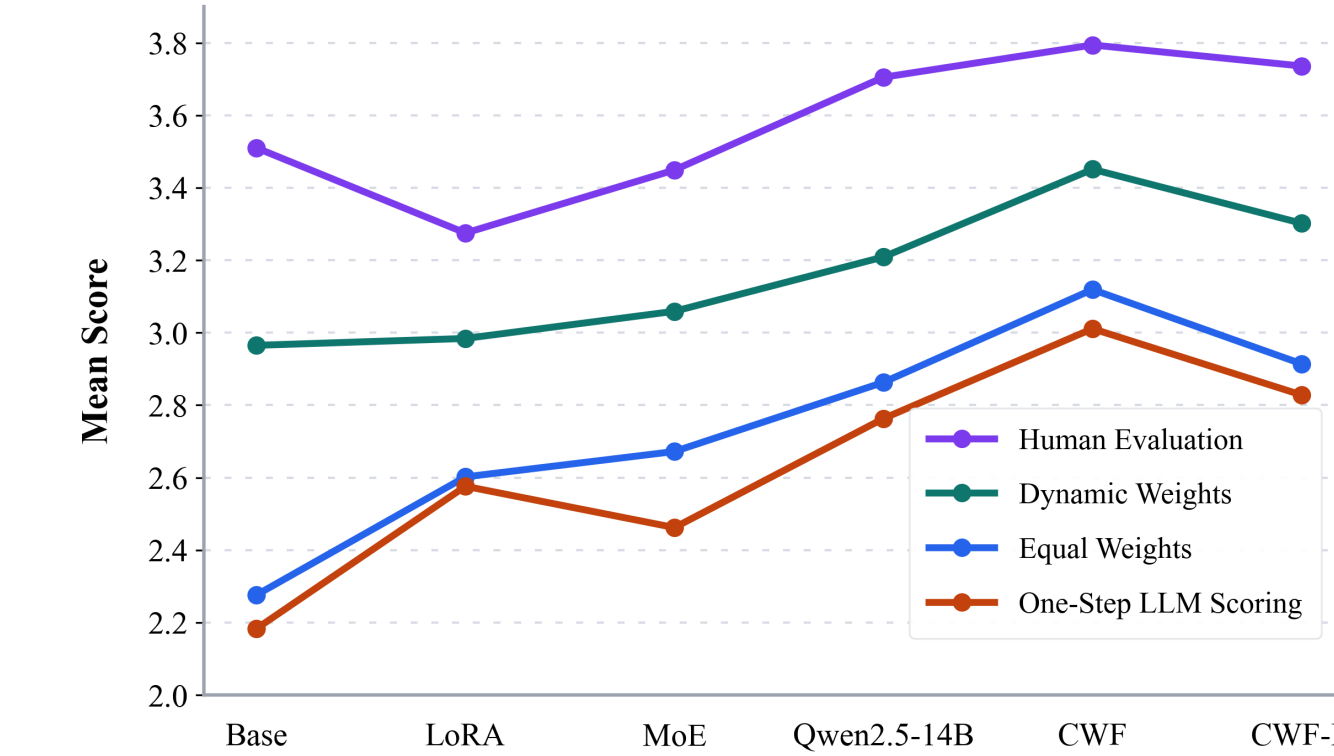
Evidence nodes have confidence 5; claims and agent statements start at the neutral prior 2.5. Signed stances $\{-1, -0.2, 0, 0.2, 1\}$ encode opposition, neutrality and support. Confidence propagates bottom-up on a 0–5 scale through evidence and reasoning statements. The graph makes judgments traceable and supplements scarce direct evidence through indirect reasoning. A summarizer revises low-confidence or inconsistent claims and merges them into the verified article. **CWF** denotes the generator; **CWF-R** adds fact-checking and revision.

Experiments and analysis

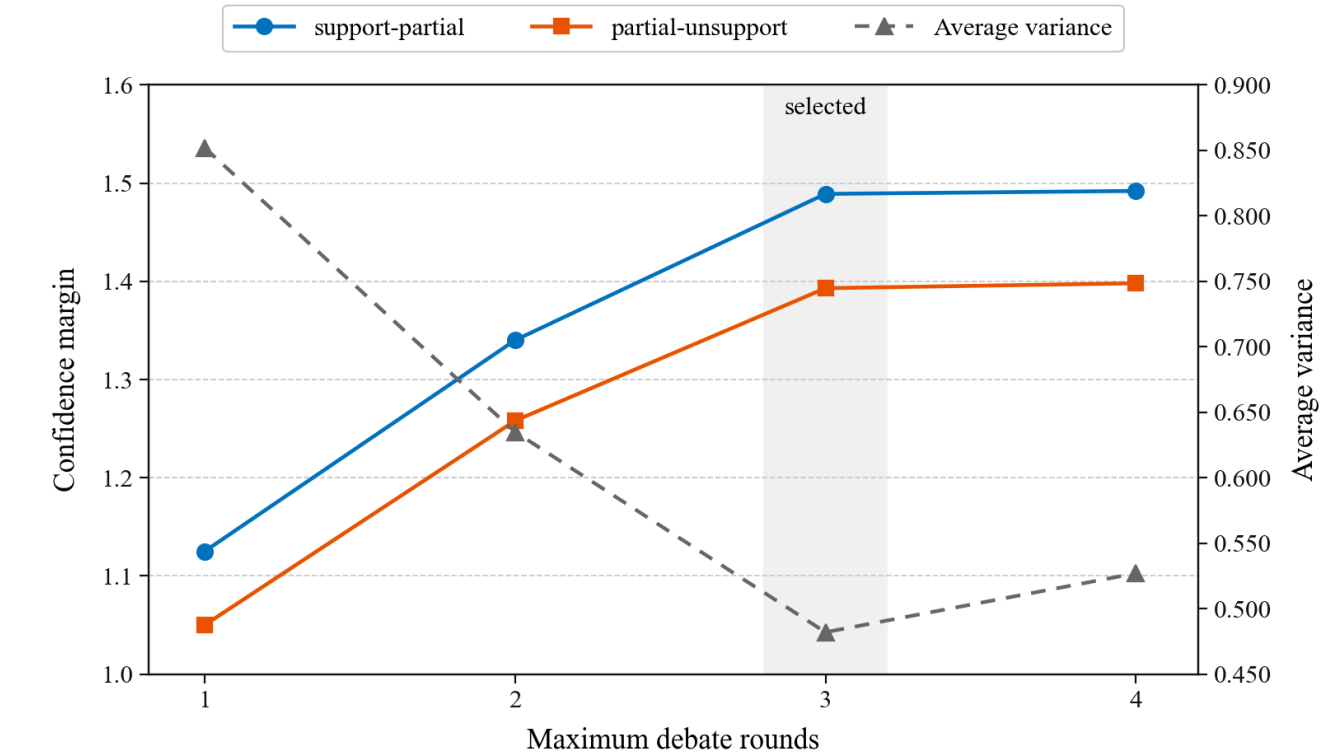


Group	Method	LLM	User	Fact. Acc.
Child	Base	2.776	3.293	3.18
	LoRA	2.955	3.286	3.31
	MoE	3.185	3.391	3.47
	Qwen2.5-14B	3.606	3.535	3.67
	CWF	3.683	3.542	3.72
Teens	CWF _R	3.330	3.480	4.05
	Base	2.770	3.461	3.24
	LoRA	2.782	3.290	3.37
	MoE	2.869	3.355	3.52
	Qwen2.5-14B	2.829	3.680	3.73
Adult	CWF	3.066	3.845	3.88
	CWF _R	3.020	3.783	4.21
	Base	3.442	3.775	3.31
	LoRA	3.218	3.248	3.44
	MoE	3.124	3.601	3.57
	Qwen2.5-14B	3.192	3.900	3.79
	CWF	3.603	3.994	3.93
	CWF _R	3.457	3.944	4.26

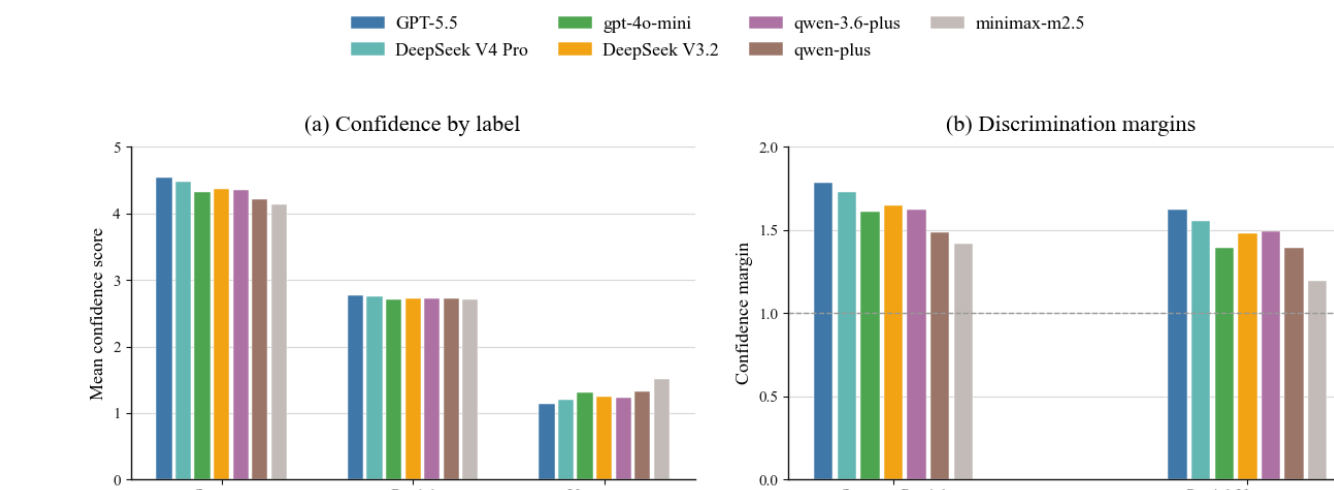
Table 4. Complete six-system, three-audience results. LLM/User: personalization; Fact. Acc.: article-level graph score. All scores: 0–5 (original, p. 8).



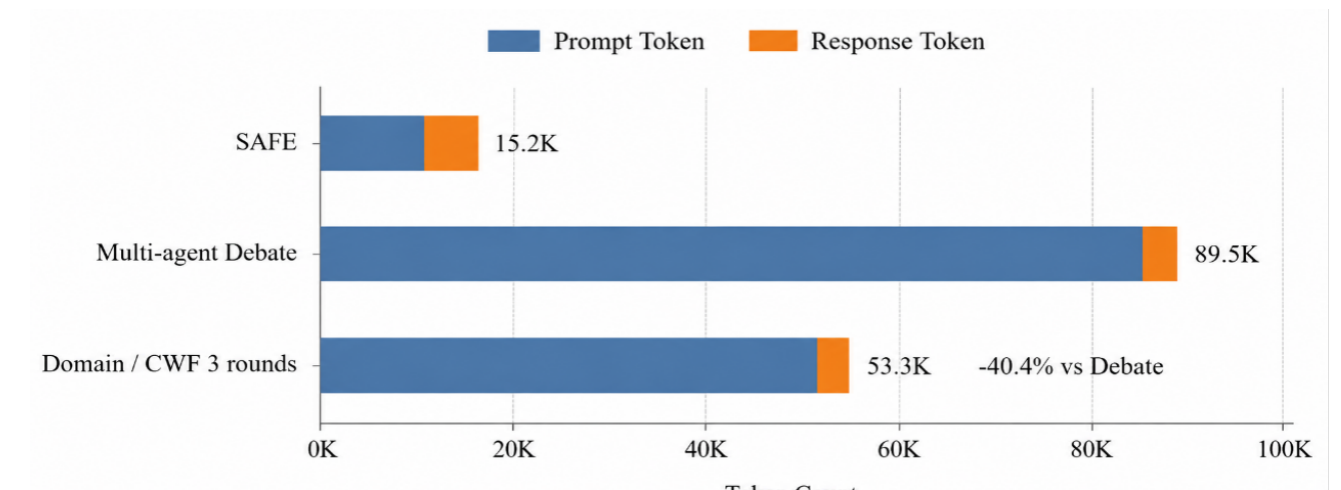
(a) Figure 3. Human–LLM consistency (p. 7).



(b) Figure 4. Debate rounds (p. 14).



(c) Figure 5. Backbone robustness (p. 15).



(d) Figure 6. Token usage (p. 16).

Setup and main gains. Base is Qwen2.5-3B; Qwen2.5-14B is the larger baseline. LoRA, MoE, CWF and CWF-R use the 3B backbone. Relative to Base, CWF improves average LLM personalization by **15.2%** and human ratings by **8.1%**; CWF-R improves factual accuracy by **28.7%**.

Verification analysis. Without debate, the supported–partial confidence gap falls from 1.489 to 1.061. Without graph propagation, unsupported-claim confidence rises from 1.327 to 1.742. SAFE F1@64 increases from 0.577 (CWF) to 0.717 (CWF-R), on a 0–1 scale. Three rounds balance discrimination and stability; a fourth gives small gains and higher variance. Figure 5 compares supported, partially supported and unsupported claims across backbone models, exposing both confidence separation and model sensitivity.

Cost and limits. Figure 6 reports 53.3K tokens for CWF versus 89.5K for multi-agent debate (40.4% fewer). The reported percentage improvements are relative gains over Base, while confidence scores and SAFE ratios use different scales. Revision can lower audience fit, especially for children. Coverage remains limited to three scientific domains and text-only personalization.

Selected references

- [1] Li et al. (2024). MixLoRA: Enhancing large language models fine-tuning with LoRA-based mixture of experts.
- [2] Min et al. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long-form text generation.
- [3] Sweller (1988). Cognitive load during problem solving: Effects on learning.
- [4] Wei et al. (2024). Long-form factuality in large language models.

